fn make_stable_truncate

Michael Shoemate (@Shoeboxam)

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `make_stable_truncate` in `mod.rs` at commit a105afe (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

None

Caller-verified

None

Pseudocode

```
def make_stable_truncate(
    input_domain: DslPlanDomain,
    input_metric: FrameDistance[SymmetricIdDistance],
    plan: DslPlan,
) -> Transformation[
    DslPlanDomain,
    DslPlanDomain,
    FrameDistance[SymmetricIdDistance],
    FrameDistance[SymmetricDistance],
]:
    # the identifier is protected from changes, so we can use the identifier from the input metric
    # instead of the identifier from the middle_metric to match truncations
    input, truncations, truncation_bounds = match_truncations(
        plan, input_metric[0].identifier
    )

    if truncations.is_empty():
        return ValueError("failed to match truncation")

    t_prior = input.make_stable(input_domain, input_metric)
    middle_domain, middle_metric = t_prior.output_space()

    for bound in truncation_bounds:
        for key in bound.by:
```

[1] See new changes with `git diff a105afe..e28f9c7 rust/src/transformations/make_stable_lazyframe/truncate/mod.rs`

```
            # raises if the key is not infallible row-by-row
            make_stable_expr(  #
                WildExprDomain(
                    columns=middle_domain.series_domains,
                    context=Context.RowByRow,
                ),
                L0PInfDistance(middle_metric[0]),
                key,
            )

    output_domain = middle_domain.clone()
    for truncation in truncations:  #
        output_domain = truncate_domain(output_domain, truncation)

    # The matcher permits only filters followed by an optional final group-by.
    has_group_by = isinstance(truncations[-1], Truncation.GroupBy)
    output_bys = [  #
        materialize_grouping_keys(bound.by) if has_group_by else bound.by
        for bound in truncation_bounds
    ]

    def function(plan: DslPlan) -> DslPlan:
        for truncation in truncations:
            match truncation:
                case Truncation.Filter(predicate):
                    plan = DslPlan.Filter(
                        input=plan,
                        predicate=predicate,
                    )

                case Truncation.GroupBy(keys, aggs):
                    plan = DslPlan.GroupBy(
                        input=plan,
                        keys=keys,
                        aggs=aggs,
                        apply=None,
                        maintain_order=False,
                        options=GroupbyOptions.default(),
                    )
        return plan

    def stability_map(id_bounds: Bounds) -> Bounds:
        #
        total_num_ids = id_bounds.get_bound({}).per_group

        # each truncation is used to derive row bounds
        new_bounds = []
        for truncation_bound, output_by in zip(  #
            truncation_bounds, output_bys
        ):
            # Look up the per-id bound using the original input expression.
            row_bound = truncate_id_bound(  #
                id_bounds.get_bound(truncation_bound.by),  #
                truncation_bound,
                total_num_ids,
            )
            # Describe the resulting rows using materialized output columns.
            row_bound.by = output_by  #
            new_bounds.append(row_bound)
        return Bounds(new_bounds)

    t_truncate = Transformation.new(
        middle_domain,
        output_domain,
```

```
        Function.new(function),
        middle_metric,
        FrameDistance(SymmetricDistance),
        StabilityMap.new_fallible(stability_map),
    )
    return t_prior >> t_truncate
```

Postcondition

Theorem 1.1. For every setting of the input parameters `(input_domain, input_metric, plan)` to `make_stable_truncate` such that the given preconditions hold,
`make_stable_truncate` raises an error (at compile time or run time) or returns a valid transformation. A valid transformation has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Appropriate output domain). For every member x in `input_domain`, `function`(x) is in `output_domain` or raises a data-independent runtime error.

3. (Stability guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
`output_metric`, if x, x' are `d_in`-close under `input_metric`, `stability_map(d_in)` does not raise an error, and `stability_map(d_in)` $=$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_metric`.

Appropriate Output Domain. By line 26, the grouping keys are stable row-by-row transformations of the data, therefore the preconditions of 36 are satisfied. By the postcondition of 36, for every element x in `input_domain`, `function`(x) is in `output_domain` or raises a data-independent runtime exception. □

Stability guarantee. By line 67, `total_num_ids` is the total number of ids an individual may contribute to a dataset:

$$d_{\text{SymId}}(\texttt{function}(x), \texttt{function}(x')) \leq \texttt{total_num_ids}. \tag{1}$$

By the postcondition of `match_truncations`, for each `truncation_bound` on line 72,

$$\max_{id} ||d_{\text{Sym}}(\texttt{function}(x)_{id,g}, \texttt{function}(x')_{id,g})||_\infty \leq \texttt{truncation.per_group},$$
$$\max_{id} ||d_{\text{Sym}}(\texttt{function}(x)_{id,g}, \texttt{function}(x')_{id,g})||_0 \leq \texttt{truncation.num_groups},$$

where g labels a group defined by the original input expressions in `truncation_bound.by`. If a final groupby is present, these labels are stored in its materialized output columns. They must not be obtained by re-evaluating the original expressions after the aggregation.

The preconditions of 76 are satisfied on line 77 (`id_bound.by` is equal to `truncation.by`), so by the postcondition of `truncate_id_bound`,

$$||d_{\text{Sym}}(\texttt{function}(x)_g, \texttt{function}(x')_g)||_\infty \leq \texttt{row_bound.per_group},$$
$$||d_{\text{Sym}}(\texttt{function}(x)_g, \texttt{function}(x')_g)||_0 \leq \texttt{row_bound.num_groups},$$

where `row_bound` denotes the return value.

Input bound lookup on line 77 uses the original expressions, so that `truncate_id_bound` receives matching input grouping keys. Line 41 computes the output coordinate system: when a final groupby is present, each original key is replaced by `col(output_name)`; for filter-only truncations, the keys are unchanged. The matcher permits only filters followed by an optional final groupby, and requires each preceding filter's keys

to be a subset of that groupby's keys. Thus all translated keys refer to values materialized by that same final groupby. Line 82 assigns those output keys only after the original input bound has been used. This preserves the numerical bounds while expressing them for the output rows.

For each truncation on line 72, `truncate_id_bound` on line 76 computes upper bounds on the resulting distance between adjacent datasets. All acquired bounds are valid upper bounds on the distance between the two datasets, by the postcondition of `match_truncations`, that each truncation does not invalidate the truncation bounds of the previous truncations.

It is shown that for every pair of elements x, x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`, if x, x' are `d_in`-close under `input_metric`, `stability_map`(`d_in`) does not raise an exception, and `stability_map`(`d_in`) $\leq$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_metric`. □